

Crawlspace Condensation

Q: I was called to a doublewide mobile home to investigate a problem with a 4-year-old duct system (the original duct system was no longer being used). Looking down the supply registers, I noticed moisture between the inner liner and the insulation. As I crawled into the crawlspace, I saw that the flex ducts were soaked. The condensation was “raining” off the duct. A previous contractor had even poked a hole in a duct to drain out the water. The duct system should be replaced, but that will not fix the problem. There are outside air vents into the crawlspace. How do I fix the problem, and not just the ducts?

—Raining Ducts in Terre Haute, Indiana

A: Whenever there is a moisture problem anywhere in the home, you have to ask, “Where did the moisture come from?” You seem to suspect that the crawlspace vents may be involved, and I bet they are. These vents

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are not removing moisture from the crawlspace. They are adding moisture.

The outdoor air in Terre Haute has a dew point above 64°F for more than 1,000 hours during the summer. Any surface whose temperature is below the dew point will condense water. Since the supply air flowing through the ducts is at 57°F or less, the duct liner and all the connections (including the boots) are easily at 64°F or less. When the outside air contacts the cooler boots, water condenses and runs down inside the duct between the liner and the insulation. You don't have a chance at preventing this condensation from happening unless you keep outside air away from any of these cool surfaces.

Incidentally, Terre Haute is not unique in this respect. Any hot climate with significant humidity has the problem of the outside air often having a higher dew point than the duct and boot surfaces. The more hours in a given year that a city experiences a dew

point above 64°F, the more likely it is that duct surface condensation will occur (see Table 1).

The best initial approach to preventing condensation is to seal the walls and floor of the crawlspace and eliminate the crawlspace ventilation (see Figure 1). That's right, close off the outside air from the crawlspace (see “Drying Out a Crawlspace,” *HE* Jan/Feb '00, p. 39). Once that is done, the duct system should be replaced, and the outside vapor retarder/air barrier should be made watertight. You don't want air to get to the cold connections. Here are the specific steps you need to take:

- Seal the duct inner liner to the boot and

takeoff with mastic and mesh.

- Seal the boot to the floor and the takeoff to the trunk with mastic and mesh.
 - Pull the insulation over the joint and then pull the outer vapor retarder/air barrier over the insulation. Make sure none of the insulation sticks out.
 - Seal the outer vapor retarder to the boot with mastic and mesh. This way, any moisture on the boot will run to the outside of the outer vapor retarder.
 - Insulate the boot and other connections with a vapor retarder on the outside.
 - Seal the top and sides of the boot vapor retarder to the floor or to the boot itself.
- Experts disagree over the best way to

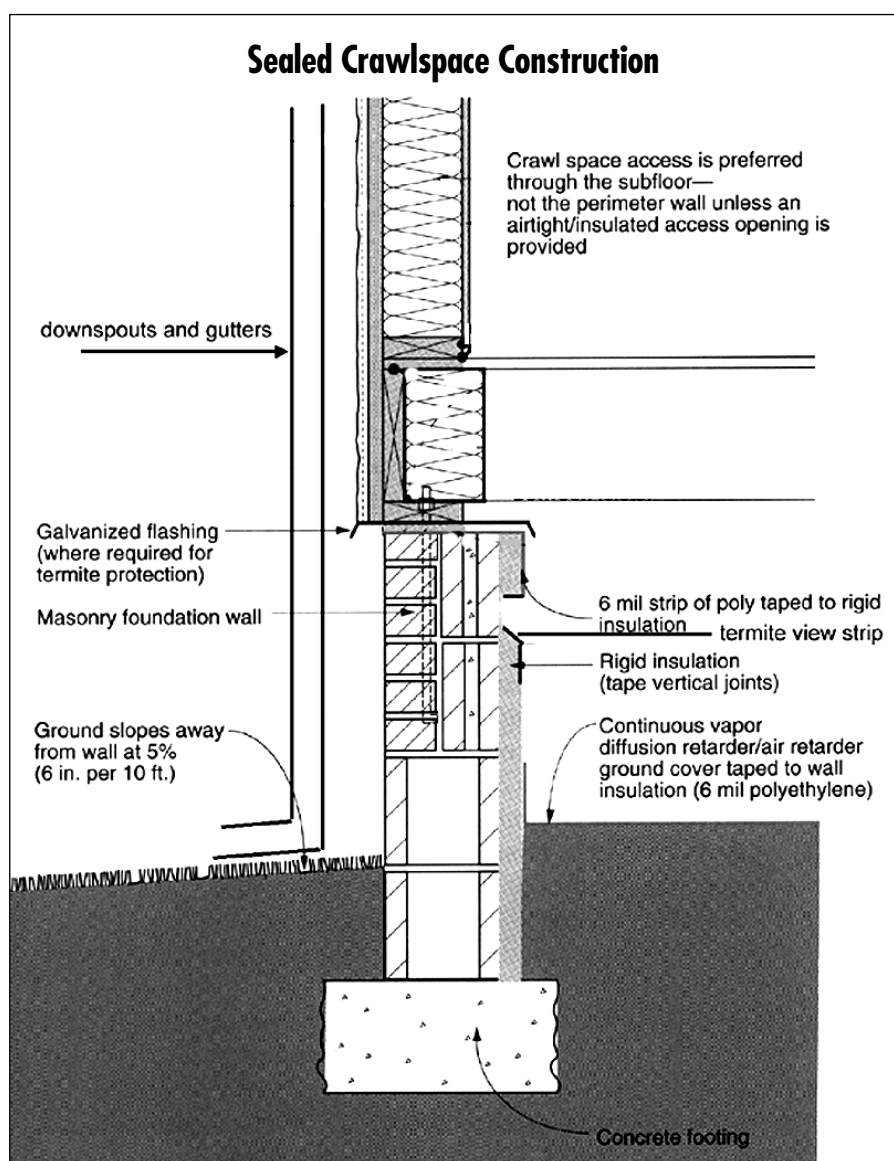


Figure 1. To keep moisture out of the crawlspace, the floor must be lined with a continuous vapor barrier (overlapping seams that are sealed), a break must be present in the lining, on the wall, to prevent termites from getting to the wood, and all crawlspace vents should be sealed off.

Table 1. Hot, Humid Cities

Dew Point Above 64°F for More than 2,000 Hours	Dew Point Above 64°F for More than 1,500 Hours	Dew Point Above 64°F for More than 1,000 Hours
Birmingham, Alabama	Honolulu, Hawaii	Wilmington, Delaware
Huntsville, Alabama	Topeka, Kansas	Hilo, Hawaii
Mobile, Alabama	Wichita, Kansas	Peoria, Illinois
Little Rock, Arkansas	Louisville, Kentucky	Indianapolis, Indiana
Jacksonville, Florida	Shreveport, Louisiana	South Bend, Indiana
Miami, Florida	Meridian, Mississippi	Sioux City, Iowa
Tampa Bay, Florida	Kansas City, Missouri	Dodge City, Kansas
Atlanta, Georgia	St. Louis, Missouri	Lexington, Kentucky
Augusta, Georgia	Omaha, Nebraska	New Orleans, Louisiana
Macon, Georgia	Charlotte, North Carolina	Baltimore, Maryland
Baton Rouge, Louisiana	Oklahoma City, Oklahoma	Grand Island, Nebraska
Jackson, Mississippi	Tulsa, Oklahoma	Newark, New Jersey
Atlantic City, New Jersey	Harrisburg, Pennsylvania	Greensboro, North Carolina
Raleigh, North Carolina	Charleston, West Virginia	Charleston, South Carolina
Philadelphia, Pennsylvania	Huntington, West Virginia	Roanoke, Virginia

execute the last step. Either seal the bottom edge of the boot vapor retarder (which could trap any condensation) or leave the bottom edge open (which would let any condensation run out). I lean toward leaving the bottom edge open and creating a type of open condensate screen.

You may have other challenges, such as:

- plumbing leaks;
- ground water (Terre Haute is on the Wabash River.);
- a depressurized crawlspace when the air handler is on; or
- problems with rainwater runoff.

Interestingly, low air flow through the air conditioner will make the situation worse by lowering the duct temperatures. Mistake to avoid: Make sure the crawlspace is dry before you seal it.



Thanks to John Tooley and Bruce Davis of Advanced Energy for their insights on this issue.

*This column states the views of the author, not necessarily those of **Home Energy** magazine. Send your comments or questions about HVAC systems and their interactions with the house as a whole to: proctor@homeenergy.org.*